Single Center Problem in Location Optimization: Model and Implementation with Python

April 3, 2025

1 Example Testing

1.1 Package

Declare the package path contained:

[325]:
```
import sys
sys.path.append('ipynb/location/packages')
```

1.1.1 Graph Visualization

Source referred: NetworkX

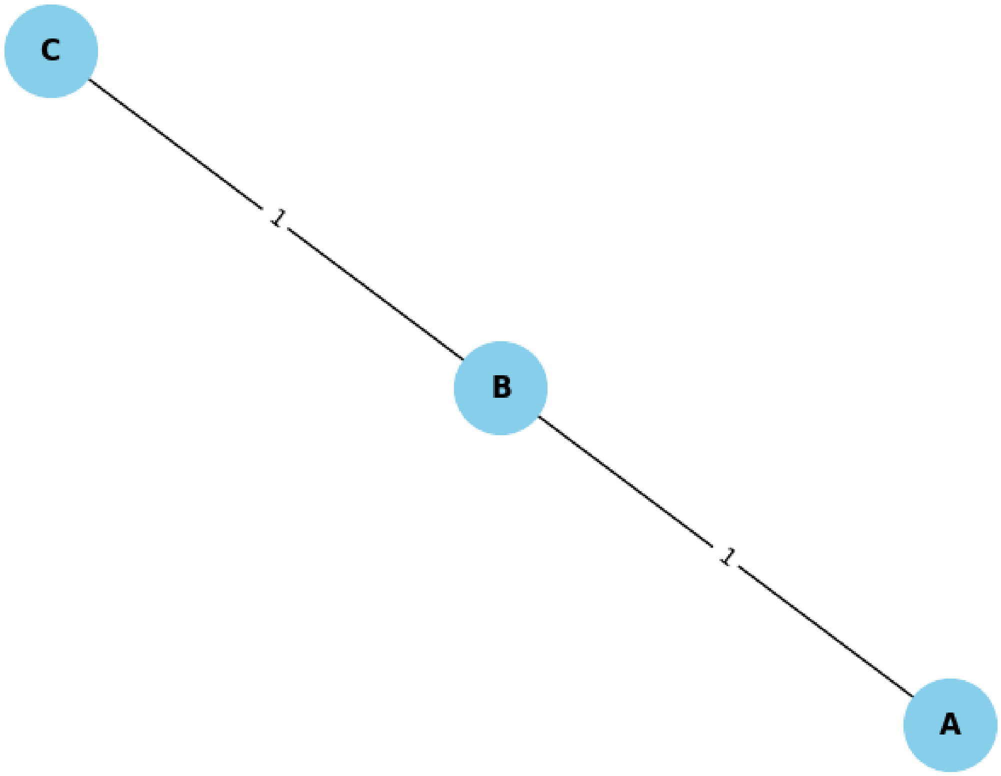
Declaring

[326]:
```
from packages.GraphView import Graph
```

Example

[327]:
```
matrix = [
    [0, 1, 0],
    [1, 0, 1],
    [0, 1, 0]
]

G = Graph(matrix)
G.show()
```



1.1.2 Shortest-path

Source referred: GeeksforGeeks

Declaring

[328]:
```
from packages import ShortestPath
```

Example

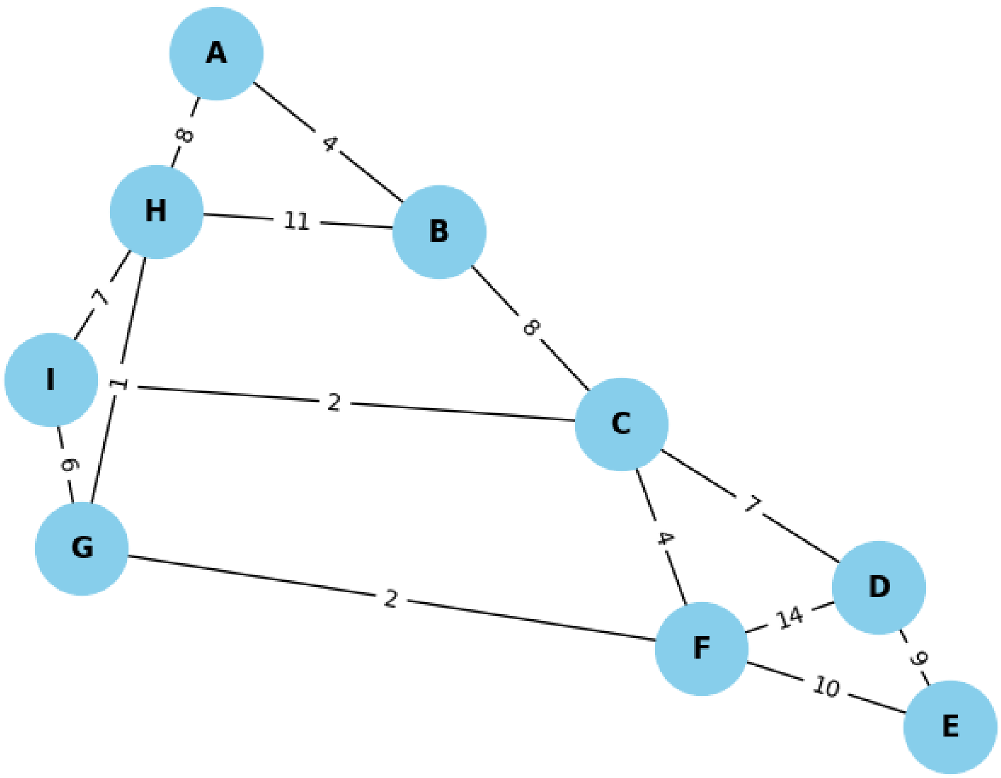
Graph parameter

[329]:
```
g = ShortestPath.Graph(9)
g.graph = [[0, 4, 0, 0, 0, 0, 0, 8, 0],
           [4, 0, 8, 0, 0, 0, 0, 11, 0],
           [0, 8, 0, 7, 0, 4, 0, 0, 2],
           [0, 0, 7, 0, 9, 14, 0, 0, 0],
           [0, 0, 0, 9, 0, 10, 0, 0, 0],
           [0, 0, 4, 14, 10, 0, 2, 0, 0],
           [0, 0, 0, 0, 0, 2, 0, 1, 6],
           [8, 11, 0, 0, 0, 0, 1, 0, 7],
           [0, 0, 2, 0, 0, 0, 6, 7, 0]
```

```
            ]
```

Visualization

[330]:
```
Visualization = Graph(g.graph)
Visualization.show()
```



Input 2 point requested

[331]:
```
start = 'D'
end = 'A'
```

Execute

[332]:
```
startInput = ord(start) - 65
endInput = ord(end) - 65
path, distance = g.getShortestPath(startInput, endInput)
result = [chr(65 + i) for i in path]
print(result)
print("Total distance:", distance)
```

```
['D', 'C', 'B', 'A']
Total distance: 19
```

1.2 Bottleneck point

1.2.1 Description

For $v \in V,\ x = (e, \alpha) \in G$, the bottleneck point v on $e = (v_i, v_j)$ is

$$\alpha l_e + d(v_i, v) = (1 - \alpha) l_e + d(v_j, v)$$

- BP_e: Set of all bottleneck points on $e \in E$.
- BP: Set of all bottleneck points on G.

1.2.2 Running

Declaring

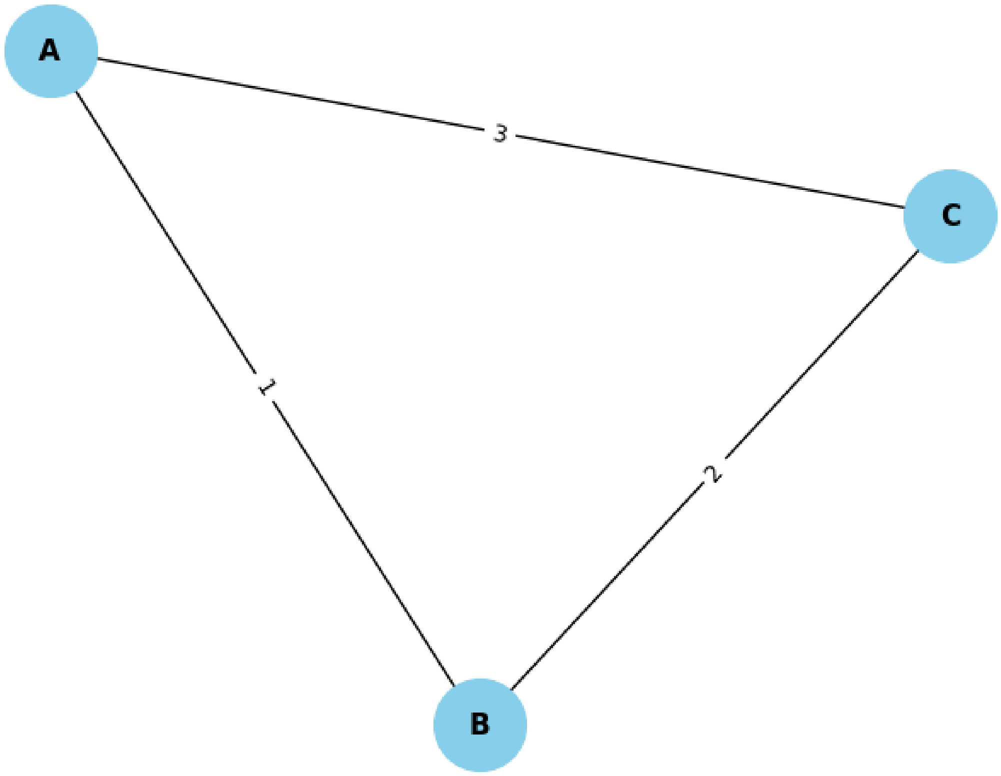
[333]:
```
from packages.Bottleneck import BP
```

Graph

[334]:
```
graph = [
    [0, 1, 3],
    [1, 0, 2],
    [3, 2, 0]
]
```

Show the graph

[335]:
```
ShowGraph = Graph(graph)
ShowGraph.show()
```



Executing

[336]:
```
a = BP(graph, 3, 'A', 'C', 'B')
print(a.alpha())
```

```
[2/3]
```

1.3 Equilibrium point

1.3.1 Description

For $v_k, v_l \in V,\ v_k \neq v_l,\ x = (e, \alpha) \in G,\ e = (v_u, v_v)$, the equililbrium point is

$$\omega_k(\alpha l_e + d(v_k, v_u)) = \omega_l((1-\alpha)l_e + d(v_l, v_v))$$

If $x \in G$

$$\omega_k d(v_k, x) = \omega_l d(v_l, x)$$

- EQ_e: Set of all equilibrium points on $e \in E$.
- EQ: Set of all equilibrium points on G.

1.4 Local center

1.4.1 Description

For $e \in E,\ e = (v_i, v_j)$, the local center on e is

$$\arg\min_{x \in e} f(x) = \arg\min\{\min_{x \in EQ_e} f(x), f(v_i), f(v_j)\}$$

1.5 Global center